

stat_conformal_overlap_frontier (v1)

TL;DR. Within a fixed, externally supplied finite list of conditionally normalized density ratios, robust weighted conformal prediction has finite-sample candidate-wise marginal coverage, and a singleton list reduces to known-ratio weighted conformal prediction. Normalization and the heavy-tail certificate give the computable CDF bound $\min\{1, K - 1, H(C, \gamma), W/2\}$. That bound does not imply a vanishing connected-length price: the universal upper bound is one, the price is zero for an effectively singleton list, and under the formally defined divisible-intervention-support condition SR_∞ the worst case over admissible support-rich fixed certificates has supremum one even with arbitrarily small CDF discrepancy. The marginal high-probability regret is zero for every n when $\eta \geq 1 - \alpha$ and eventually zero for every fixed $\eta \in (0, 1)$, via a measurable closed-hull randomized repair that needs no second moment.

1 Project justification

Gap. Known-ratio weighted conformal prediction, robust compatible-law coverage, doubly robust asymptotic calibration, and generic conformal-length optimization do not determine the population connected-length behavior or the marginal high-probability regret of this externally fixed finite normalized ratio-list model. The missing distinction is between a universal upper bound, which needs no support richness, and an order-one lower witness, which uses the explicitly defined divisible-intervention-support condition SR_∞ .

Niche. The analysis is conditional on a fixed externally supplied finite list satisfying exact target-ratio containment, candidate normalization, graded width, and a heavy upper-tail envelope. The order-one population lower conclusion is a worst case over admissible support-rich certificates under SR_∞ , not an unconditional claim for every treatment support. No procedure for learning or certifying the list is supplied.

Fill. Normalization yields the half-total-variation algebra and, after tail integration, the explicit CDF envelope $\min\{1, K - 1, H(C, \gamma), W/2\}$. The consolidated divisible-support witness shows separately that ordinary-cell CDF discrepancy has order δ , tail-shell discrepancy has order $\epsilon^{\gamma-1}$, and under SR_∞ a fixed two-candidate certificate can have connected-length-price supremum one despite arbitrarily small CDF discrepancy. The unconditional population conclusion remains only $0 \leq \text{len}(I_{\text{rob}}) - \text{len}(I_w) \leq 1$. At sample level, the criterion has an $\eta = 1 - \alpha$ elbow and an eventual exact zero-regret repair; $n^{-(\gamma-1)/\gamma}$ is sufficient construction slack rather than a positive regret exponent.

2 Related work

Tibshirani et al. provide the known-ratio weighted-conformal benchmark, while Jin–Ren–Candes provide robust compatible-law conformal baselines. Yang–Kuchibhotla–Tchetgen Tchetgen study doubly robust asymptotic calibration, and Kiyani–Pappas–Hassani study generic conformal length optimization. Qiu–Dobriban–Tchetgen Tchetgen establish finite-sample uninformative-set phenomena under unknown covariate shift, an information regime different from the present externally fixed exact-list regime. The present contribution is a finite-list diagnostic: an unconditional computable CDF bound and unit length upper bound, an order-one support-rich lower obstruction under SR_∞ , and a degeneracy result for the stated marginal regret. It claims no global superiority over those methods and no unconditional supremum-one result on arbitrary treatment supports.

3 Setup

Target (estimand / structural parameter): $I_{\text{rob}}=[m-r_{\text{rob}},m+r_{\text{rob}}] \text{ intersect } [0,1]$. Identifying functional / recovery map: $r_{\text{rob}}=\inf\{r \text{ in } [0,1]:\inf\{v \text{ in } V_{\text{epsilon}}\}E\text{Pobs}[v(X,A)1\{|Y-m|\leq r\}]\geq 1-\alpha\}$.

- n — integer; N ; index := number of calibration observations
- i — integer; $\{1, \dots, n\}$; bound index := calibration-observation index
- K — integer; N ; certificate index := number of externally certified candidate ratios
- k — integer; $\{1, \dots, K\}$; bound index := candidate-ratio index
- α — scalar; $(0,1)$; constant := target miscoverage probability
- γ — scalar; $(1,2)$; constant := upper-endpoint tail index
- ϵ — scalar; $(0,1/2)$; constant := inverse high-ratio audit threshold
- δ — scalar; $[0,1]$; constant := certified ordinary-cell ratio-width tolerance
- C — scalar; $(0,\text{infinity})$; constant := upper-endpoint tail-envelope constant
- c_0 — scalar; $(0,\text{infinity})$; constant := local score-quantile margin constant
- η — scalar; $(0,1)$; constant := outer regret probability
- $\mathcal{X}$ — measurable space; primitive space := covariate space

- $\mathcal{A}$ — measurable space; primitive space := treatment space with a dominating measure
- X — random vector; $\mathcal{X}$; full-data primitive := baseline covariate
- x — point; $\mathcal{X}$; bound variable := generic covariate value
- A — random variable; $\mathcal{A}$; full-data primitive := observed treatment
- a — point; $\mathcal{A}$; bound variable := generic treatment value
- B — set; $\text{Borel}([0,1])$; bound variable := generic Borel outcome event
- y — scalar; $[0,1]$; bound variable := candidate outcome value
- r — scalar; $[0,1]$; bound variable := candidate score radius
- $Y(a)$ — random variable; $[0,1]$; full-data primitive := potential outcome under treatment a
- $Y(A)$ — random variable; $[0,1]$; full-data primitive := potential outcome at the realized treatment
- Y — random variable; $[0,1]$; observed-data primitive := observed outcome
- Z — random element; $\mathcal{X} \times \mathcal{A} \times [0,1]^{\mathcal{A}}$; full-data unit := $Z=(X,A,(Y(a))_{a \in \mathcal{A}})$
- P_F — probability law; $\text{Prob}(\mathcal{X} \times \mathcal{A} \times [0,1]^{\mathcal{A}})$; source law := source full-data law of Z
- P_{obs} — probability law; $\text{Prob}(\mathcal{X} \times \mathcal{A} \times [0,1])$; observed-data law := law under P_F of the observed tuple $(X,A,Y(A))$
- P_X — probability law; $\text{Prob}(\mathcal{X})$; covariate law := X -marginal of P_F
- π — Markov kernel; $\mathcal{A} \times \mathcal{X} \rightarrow [0,\infty)$; GPS := $\pi(a|x)$, the P_F -conditional treatment density of A given $X=x$
- g — Markov kernel; $\mathcal{A} \times \mathcal{X} \rightarrow [0,\infty)$; intervention := $g(a|x)$, the prespecified stochastic-intervention density
- w — measurable function; $\mathcal{X} \times \mathcal{A} \rightarrow [0,\infty)$; true intervention ratio := $w(x,a)=g(a|x)/\pi(a|x)$ on $\{\pi(a|x)>0\}$
- ℓ — measurable function; $\mathcal{X} \times \mathcal{A} \rightarrow [0,\infty)$; certificate input := certified lower endpoint for w

- u — measurable function; $\mathcal{X} \times \mathcal{A} \rightarrow [0, \infty)$;
certificate input := certified upper endpoint for w
- t — scalar; $[1, \infty)$; bound variable := tail argument
- m — scalar; $[0, 1]$; score input := score centre fitted on data
independent of calibration and target draws
- S — measurable function; $[0, 1] \rightarrow [0, 1]$; nested nonconformity
score := $S(y) = |y - m|$
- O — random vector; $\mathcal{X} \times \mathcal{A} \times [0, 1]$; observed unit
:= $O = (X, A, Y)$ under P_{obs}
- O_i — random vector; $\mathcal{X} \times \mathcal{A} \times [0, 1]$; sample unit
:= i -th calibration draw from P_{obs}
- X_i — random vector; $\mathcal{X}$; sample component := covariate component
of O_i
- A_i — random variable; $\mathcal{A}$; sample component := treatment
component of O_i
- Y_i — random variable; $[0, 1]$; sample component := outcome component
of O_i
- D_n — random sample; $(\mathcal{X} \times \mathcal{A} \times [0, 1])^n$; calibration
sample := $D_n = (O_1, \dots, O_n)$
- v^k — measurable function; $\mathcal{X} \times \mathcal{A} \rightarrow [0, \infty)$;
certificate input := k -th externally certified candidate ratio
- V_{epsilon} — finite set of functions; $\{v: \mathcal{X} \times \mathcal{A} \rightarrow [0, \infty)\}$; identified set := causally compatible ratio list
from `def:compatible-ratios`
- v — measurable function; $\mathcal{X} \times \mathcal{A} \rightarrow [0, \infty)$;
bound variable := generic member of V_{epsilon}
- Q_v — probability law; $\text{Prob}(\mathcal{X} \times \mathcal{A} \times [0, 1]^{\mathcal{A}})$;
compatible intervention full-data law := law with $dQ_v/dP_F = v(X, A)$
- T_v — random pair; $\mathcal{X} \times \mathcal{A}$; target draw := candidate-specific
target covariate-treatment pair under Q_v
- U — random variable; $\text{Unif}(0, 1)$; randomization := U is $\text{Uniform}(0, 1)$
and, for every v in V_{epsilon} , is independent of the joint element
consisting of D_n , T_v , and the Q_v -target outcome $Y(A)$. The formal
complete-target statement is `ass:uniform-tie`.
- Y^v — random variable; $[0, 1]$; compatible intervention outcome
:= $Y(A)$ under Q_v

- F_v — distribution function; $[0,1] \rightarrow [0,1]$; compatible score CDF $:= F_v(r) = \mathbb{E} \text{Pobs}[v(X,A) \mathbf{1}\{S(Y) \leq r\}]$
- F_w — distribution function; $[0,1] \rightarrow [0,1]$; target score CDF $:= F_w(r) = \mathbb{E} \text{Pobs}[w(X,A) \mathbf{1}\{S(Y) \leq r\}]$
- r_v — scalar; $[0,1]$; compatible oracle radius $:= r_v = \inf\{r \text{ in } [0,1] : F_v(r) \geq 1 - \alpha\}$
- r_w — scalar; $[0,1]$; target oracle radius $:= r_w = \inf\{r \text{ in } [0,1] : F_w(r) \geq 1 - \alpha\}$
- r_{rob} — scalar; $[0,1]$; robust oracle radius $:= r_{\text{rob}} = \inf\{r \text{ in } [0,1] : \inf_{v \text{ in } V_{\text{epsilon}}} F_v(r) \geq 1 - \alpha\}$
- I_w — interval; subsets of $[0,1]$; point-ratio oracle interval $:= I_w = [m - r_w, m + r_w] \text{ intersect } [0,1]$
- I_{rob} — interval; subsets of $[0,1]$; robust oracle interval $:= I_{\text{rob}} = [m - r_{\text{rob}}, m + r_{\text{rob}}] \text{ intersect } [0,1]$
- p_{rob} — random function; $[0,1] \rightarrow [0,1]$; calibration functional $:=$ robust randomized weighted-conformal p-value from def:robust-wcp evaluated at a candidate-specific target pair T_v
- p_k — random function; $[0,1] \rightarrow [0,1]$; calibration functional $:=$ weighted-conformal p-value for candidate ratio v^k evaluated at a candidate-specific target pair T_v
- $I_{\text{hat},n}$ — random interval rule; $D_n \times (\text{mathcal{X}} \times \text{mathcal{A}}) \times (0,1) \rightarrow \{\text{intervals in } [0,1]\}$; robust conformal interval $:= I_{\text{hat},n}(D_n, T_v, U) = \{y \text{ in } [0,1] : p_{\text{rob}}(y) > \alpha\}$
- h_J — measurable function; $D_n \times (\text{mathcal{X}} \times \text{mathcal{A}}) \times (0,1) \rightarrow [0,1]$; bound variable $:=$ radius function defining the score-centred rule J
- J — measurable rule; $D_n \times (\text{mathcal{X}} \times \text{mathcal{A}}) \times (0,1) \rightarrow \{\text{connected intervals in } [0,1]\}$; bound variable $:= J(D_n, T_v, U) = [m - h_J(D_n, T_v, U), m + h_J(D_n, T_v, U)] \text{ intersect } [0,1]$
- len — functional; interval $\rightarrow [0,1]$; length functional $:= \text{len}(I) := \lambda_1(I)$ for every interval $I \subseteq [0,1]$, including open and half-open intervals, with $\text{len}([b, c]) = c - b$.
- $\text{mathfrak{P}_{\text{cert}}}$ — class of probability laws; subsets of $\text{Prob}(\text{mathcal{X}} \times \text{mathcal{A}} \times [0,1]^{\text{mathcal{A}}})$; model class $:=$ certified-tail full-data class from def:certificate-class
- R_n — nonnegative scalar; $[0, \text{infinity})$; resolved minimax functional $:=$ minimax high-probability robust-oracle regret from def:minimax-regret

4 Assumptions

Assumption 1 (ass:iid-source). $(O_1, \dots, O_n)$ is distributed as P_{obs}^n Standard (i.i.d. sampling, [2]).

Assumption 2 (ass:compatible-target-law). For every v in V_{epsilon} , (T_v, Y^v) has the Q_v -law of $(X, A, Y(A))$.

Standard (covariate-shift target sampling with candidate-specific target copies, [2]).

Assumption 3 (ass:independent-target). For every v in V_{epsilon} , (T_v, Y^v, U) is independent of D_n .

Standard (independent calibration and target samples, [2]).

Assumption 4 (ass:uniform-tie). U is $\text{Uniform}(0, 1)$ and independent of the joint element (D_n, T_v, Y^v) for every v in V_{epsilon} .

Standard (independent randomized conformal tie breaking, [2]).

Assumption 5 (ass:consistency). $Y = Y(A)$ almost surely under P_F

Standard (consistency, [4]).

Assumption 6 (ass:exchangeability). $Y(a)$ is independent of A conditional on X under P_F for every a in $\text{mathcal{A}}$

Standard (conditional exchangeability, [4]).

Assumption 7 (ass:intervention-domination). $g(a/x) > 0$ implies $\pi(a/x) > 0$ for P_X -almost every x and every a in $\text{mathcal{A}}$

Standard (positivity for stochastic interventions, [4]).

Assumption 8 (ass:bounded-outcome). $Y(a)$ belongs to $[0, 1]$ almost surely under P_F for every a in $\text{mathcal{A}}$

Standard (bounded outcome, [3]).

Assumption 9 (ass:candidate-normalization). For every k in $\{1, \dots, K\}$, integral $v^k(x, a) \pi(a/x) da = 1$ for P_X -almost every x .

Novel. This is an externally supplied conditional premise on the finite candidate list. No procedure for generating, estimating, or auditing the list or its normalization is asserted.

Assumption 10 (ass:certificate-containment). w belongs to V_{epsilon} .

Novel. This is an externally supplied conditional premise that the fixed finite list contains the true intervention ratio. No certification or list-learning procedure is asserted.

Assumption 11 (ass:upper-tail). $P_{\text{obs}}\{u(X, A) > t\} \leq C t^{(-\gamma)}$ for every $t \geq 1$

Novel. This Pareto envelope is an externally supplied conditional premise on the fixed ratio envelope u . No procedure for estimating or certifying it is asserted.

Assumption 12 (ass:graded-certificate-width). $0 \leq u(X, A) - \text{ell}(X, A) \leq \delta$
 $1\{u(X, A) \leq \epsilon\} + u(X, A) 1\{u(X, A) > \epsilon\}$ almost surely
under P_{obs}

Novel. This graded-width inequality is an externally supplied conditional premise on the fixed lower and upper ratio envelopes. No procedure for producing or validating those envelopes is asserted.

Assumption 13 (ass:quantile-margin). $F_w(r_w + t) - F_w(r_w - t) \geq c_0$ for every t in $[0, 1]$ for which $0 \leq r_w - t$ and $r_w + t \leq 1$

Novel. This externally supplied conditional premise requires mass across each admissible symmetric neighborhood of r_w . It permits an atom at r_w , may be one-side-vacuous at a boundary quantile, and permits a flat interval immediately to the right of r_w ; therefore it does not guarantee stable upper-quantile inversion.

5 Definitions

Definition 1 (def:compatible-ratios). Finite causally compatible ratio list

$:= V_{\text{epsilon}} = \{v^1, \dots, v^K\}$, $\text{ell}(x, a) = \min_{1 \leq k \leq K} v^k(x, a)$, and $u(x, a) = \max_{1 \leq k \leq K} v^k(x, a)$
(inputs: K, v^k, ell, u)

Definition 2 (def:robust-oracle). Robust identified-set oracle interval

$:= I_{\text{rob}} = [m - r_{\text{rob}}, m + r_{\text{rob}}] \cap [0, 1]$, where $r_{\text{rob}} = \inf\{r \in [0, 1] : \inf_{v \in V_{\text{epsilon}}} E_{P_{\text{obs}}}[v(X, A) 1\{|Y - m| \leq r\}] \geq 1 - \alpha\}$ (inputs: $P_{\text{obs}}, V_{\text{epsilon}}, m, \alpha$)

Definition 3 (def:robust-wcp). Randomized robust weighted-conformal interval

$:=$ For a candidate v in V_{epsilon} and its target pair T_v , for y in $[0, 1]$ and $k = 1, \dots, K$, set $p_k(y) = [\sum_{i=1}^n v^k(X_i, A_i) 1\{S(Y_i) > S(y)\} + U(v^k(T_v) + \sum_{i=1}^n v^k(X_i, A_i) 1\{S(Y_i) = S(y)\})] / [v^k(T_v) + \sum_{i=1}^n v^k(X_i, A_i)]$ when its denominator is positive, and set $p_k(y) = 0$ otherwise. Set $p_{\text{rob}}(y) = \max_{1 \leq k \leq K} p_k(y)$ and output $\hat{I}_{\text{hat}, n}(D_n, T_v, U) = \{y \in [0, 1] : p_{\text{rob}}(y) > \alpha\} = \bigcup_{k=1}^K \{y \in [0, 1] : p_k(y) > \alpha\}$. (inputs: $D_n, T_v, U, K, v^k, V_{\text{epsilon}}, S, \alpha$)

Definition 4 (def:certificate-class). Certified-tail full-data class $:= \text{mathfrak{P-cert}}(C, \gamma, \epsilon, \eta)$

in $\text{Prob}(\text{mathcal{X}} \times \text{mathcal{A}} \times [0, 1]^{\text{mathcal{A}}})$: ass:consistency, ass:exchangeability, ass:intervention-domination, ass:bounded-outcome, ass:candidate-normalization, ass:certificate-containment, ass:upper-tail, ass:graded-certificate-width, and ass:quantile-margin hold for the P_F -induced objects} — the class of all objects satisfying: ass:consistency; ass:exchangeability; ass:intervention-domination; ass:bounded-outcome; ass:candidate-normalization; ass:certificate-containment; ass:upper-tail; ass:graded-certificate-width; ass:quantile-margin.

Definition 5 (def:minimax-regret). High-probability robust-oracle regret

$:= R_n(\eta) = \inf_J \sup_{P_F \in \text{mathfrak{P-cert}}} \inf\{r \geq 0 : \sup_{v \in V_{\text{epsilon}}} \Pr\{D_n, T_v, U\} \leq r\}$ where J has the score-centred form declared for J and satisfies $\inf_{v \in V_{\text{epsilon}}} \Pr\{D_n, T_v, U\} \leq r$

in $V_epsilon\}Pr_{\{D_n, T_v, Y^v, U\}}\{Y^v \text{ in } J(D_n, T_v, U)\} \geq 1 - \alpha$ for every P_F in $\text{mathfrak{P_cert}}$, with each (T_v, Y^v) distributed as in *ass:compatible-target-law*.
(inputs: $n, \epsilon, \text{mathfrak{P_cert}}, I_rob, len, V_epsilon, D_n, T_v, Y^v, U, J$)

6 Main results

Proposition 1 (prop:causal-change-of-measure). *For every v in $V_epsilon$ and every B in $\text{Borel}([0, 1])$, $Q_v\{Y(A) \text{ in } B\} = E_Pobs[v(X, A)\mathbf{1}\{Y \text{ in } B\}]$. At $v=w$, this equals $E_PX[\text{integral } g(a/X)Pr_PF\{Y(a) \text{ in } B/X\} da]$.*

Proof. Let $v \in V_\epsilon$ and $B \in \mathcal{B}([0, 1])$. Write ν for the dominating measure on $\mathcal{A}$. Candidate normalization and iterated expectation give

$$\mathbb{E}_{P_F} v(X, A) = \mathbb{E}_{P_X} \left[\int_{\mathcal{A}} v(X, a) \pi(a | X) \nu(da) \right] = 1.$$

Thus the nonnegative Radon–Nikodym derivative in the definition of Q_v has total mass one. By that definition,

$$Q_v\{Y(A) \in B\} = \mathbb{E}_{P_F} [v(X, A)\mathbf{1}\{Y(A) \in B\}].$$

Consistency implies $Y = Y(A)$, P_F -almost surely. Since P_{obs} is the induced law of the observed tuple, it follows that

$$Q_v\{Y(A) \in B\} = \mathbb{E}_{P_{\text{obs}}} [v(X, A)\mathbf{1}\{Y \in B\}].$$

This proves the first assertion for every candidate ratio.

For the second assertion, substitute the true ratio w into the observable weighting functional. Consistency and disintegration with respect to X and A yield

$$\begin{aligned} \mathbb{E}_{P_{\text{obs}}} [w(X, A)\mathbf{1}\{Y \in B\}] &= \mathbb{E}_{P_F} [w(X, A)\mathbf{1}\{Y(A) \in B\}] \\ &= \mathbb{E}_{P_X} \left[\int_{\mathcal{A}} w(X, a) \pi(a | X) \Pr_{P_F}\{Y(a) \in B | X, A = a\} \nu(da) \right]. \end{aligned}$$

Conditional exchangeability gives, for P_X -almost every X and $\pi(\cdot | X)\nu$ -almost every a ,

$$\Pr_{P_F}\{Y(a) \in B | X, A = a\} = \Pr_{P_F}\{Y(a) \in B | X\}.$$

On $\{\pi(a | X) > 0\}$, the definition of w gives $w(X, a)\pi(a | X) = g(a | X)$. On $\{\pi(a | X) = 0\}$, intervention domination gives $g(a | X) = 0$, so the same equality holds after assigning any value to w there. Substitution therefore gives

$$\mathbb{E}_{P_{\text{obs}}} [w(X, A)\mathbf{1}\{Y \in B\}] = \mathbb{E}_{P_X} \left[\int_{\mathcal{A}} g(a | X) \Pr_{P_F}\{Y(a) \in B | X\} \nu(da) \right],$$

which is the claimed stochastic-intervention formula. $\square$

Justification. The proposition keeps each compatible causal target law distinct from its observed-law weighting functional and recovers the stochastic-intervention g-formula at w . *Gap.* This is the standard causal change-of-measure substrate; no identification improvement over known stochastic-intervention formulas is claimed. *Consumer.* It supports candidate-wise marginal coverage, evaluated separately under each compatible stochastic-intervention law, and does not assert a joint event across candidate laws.

Lemma 1 (lem:certificate-cdf-modulus). *For every P_F in $\mathbf{mathfrak{P}}_{\text{cert}}$ and every v in V_{epsilon} , with all expectations under P_{obs} , $\sup_{r \in [0,1]} |F_v(r) - F_w(r)| \leq (1/2)E|v-w| \leq \min\{1, E(u-w), E(w-\ell), (1/2)E(u-\ell)\}$. Let $L = \epsilon^{-1}$, $H(C, \gamma) = C/(\gamma-1)$ when $C \leq 1$ and $H(C, \gamma) = (\gamma/(\gamma-1))C^\gamma$ when $C > 1$, and $W = \delta + (L - \delta) \min\{1, CL^\gamma\} + \int_0^L L^\gamma dt$. Then $\sup_{v \in V_{\text{epsilon}}} \sup_{r \in [0,1]} |F_v(r) - F_w(r)| \leq \min\{1, K-1, H(C, \gamma), W/2\}$. Equivalently, $W = \delta + C[(\gamma/(\gamma-1))\epsilon^{(\gamma-1)} - \epsilon^\gamma]$ if $C \epsilon^{(\gamma-1)} \leq 1$, and $W = (\gamma/(\gamma-1))C^\gamma (1/\gamma)$ if $C \epsilon^{(\gamma-1)} > 1$.*

Proof. Fix $P_F \in \mathfrak{P}_{\text{cert}}$ and $v \in V_\epsilon$, let ν be the dominating measure on $\mathcal{A}$, and write $\mathbb{E}$ for expectation under P_{obs} . By candidate normalization, every v^k satisfies

$$\mathbb{E}v^k(X, A) = \mathbb{E}_{P_X} \left[\int_{\mathcal{A}} v^k(X, a) \pi(a | X) \nu(da) \right] = 1.$$

Certificate containment gives $w = v^j$, P_{obs} -almost surely, for some j , so $\mathbb{E}w = 1$ as well.

Put $D = v - w$. Since $\mathbb{E}D = 0$,

$$\mathbb{E}D_+ = \mathbb{E}D_- = \frac{1}{2}\mathbb{E}|D|.$$

For every measurable event A ,

$$-\mathbb{E}D_- \leq \mathbb{E}[D\mathbf{1}_A] \leq \mathbb{E}D_+,$$

and hence

$$|\mathbb{E}[D\mathbf{1}_A]| \leq \frac{1}{2}\mathbb{E}|D|.$$

Applying this inequality to $A = \{S(Y) \leq r\}$ and taking the supremum over $r \in [0, 1]$ proves

$$\sup_{r \in [0,1]} |F_v(r) - F_w(r)| \leq \frac{1}{2}\mathbb{E}|v - w|.$$

The envelope definition and containment imply

$$\ell \leq v, \quad w \leq u \quad P_{\text{obs}}\text{-almost surely.}$$

Consequently,

$$(v - w)_+ \leq u - w, \quad (w - v)_+ \leq w - \ell, \quad |v - w| \leq u - \ell.$$

Together with the preceding equality of the positive and negative masses, these inequalities yield

$$\frac{1}{2}\mathbb{E}|v - w| \leq \min\left\{\mathbb{E}(u - w), \mathbb{E}(w - \ell), \frac{1}{2}\mathbb{E}(u - \ell)\right\}.$$

It is also at most one, because it is the total-variation distance between two probability densities. Moreover, after choosing j with $w = v^j$,

$$u - w \leq \sum_{k \neq j} v^k,$$

so normalization gives

$$\mathbb{E}(u - w) \leq K - 1.$$

We next turn the tail condition into the bound $H(C, \gamma)$. Since $\mathbb{E}w = 1$,

$$\mathbb{E}(u - w) = \mathbb{E}u - 1.$$

Layer cake and the fact that probabilities are at most one give

$$\mathbb{E}u - 1 \leq \int_1^\infty \min\{1, Ct^{-\gamma}\} dt.$$

If $C \leq 1$, this integral is $C/(\gamma - 1)$. If $C > 1$, split it at $C^{1/\gamma}$ to obtain

$$C^{1/\gamma} - 1 + \frac{C(C^{1/\gamma})^{1-\gamma}}{\gamma - 1} = \frac{\gamma}{\gamma - 1} C^{1/\gamma} - 1.$$

Thus $\mathbb{E}(u - w) \leq H(C, \gamma)$.

Finally let $L = \epsilon^{-1}$. Because $L > 1$ and $0 \leq \delta \leq 1$, the graded-width condition and layer cake imply

$$\begin{aligned} \mathbb{E}(u - \ell) &\leq \delta \Pr(u \leq L) + \mathbb{E}[u \mathbf{1}\{u > L\}] \\ &= \delta + (L - \delta) \Pr(u > L) + \int_L^\infty \Pr(u > t) dt \\ &\leq \delta + (L - \delta) \min\{1, CL^{-\gamma}\} + \int_L^\infty \min\{1, Ct^{-\gamma}\} dt \\ &= W. \end{aligned}$$

Combining all established bounds and then taking the supremum over v proves

$$\sup_{v \in V_\epsilon} \sup_{r \in [0, 1]} |F_v(r) - F_w(r)| \leq \min\{1, K - 1, H(C, \gamma), W/2\}.$$

It remains only to evaluate W . If $C\epsilon^\gamma \leq 1$, then $CL^{-\gamma} \leq 1$ and the tail cap is inactive for every $t \geq L$. Therefore

$$\begin{aligned} W &= \delta + (L - \delta)CL^{-\gamma} + \frac{C}{\gamma - 1}L^{1-\gamma} \\ &= \delta + C \left[\frac{\gamma}{\gamma - 1}\epsilon^{\gamma-1} - \delta\epsilon^\gamma \right]. \end{aligned}$$

If $C\epsilon^\gamma > 1$, set $t_0 = C^{1/\gamma} > L$. Then

$$\begin{aligned} W &= \delta + (L - \delta) + (t_0 - L) + \frac{C}{\gamma - 1}t_0^{1-\gamma} \\ &= \frac{\gamma}{\gamma - 1}C^{1/\gamma}. \end{aligned}$$

These are the two asserted formulas. $\square$

Justification. Candidate normalization makes half total variation the exact algebraic scale, while $H(C, \gamma)$ and W turn the tail and graded-width premises into a computable envelope. *Gap.* This is a conditional finite-list certificate calculation; no joint sharpness of $\min\{1, K - 1, H(C, \gamma), W/2\}$ and no connected-length inversion follow from it alone. *Consumer.* Compute $H(C, \gamma)$ and W , then use $\min\{1, K - 1, H(C, \gamma), W/2\}$ as a CDF discrepancy bound before checking one-sided quantile growth.

Proposition 2 (prop:robust-wcp-coverage). *For every P_F in $\mathsf{mathfrak{P_cert}}$ and every v in $V_epsilon$, $\mathsf{Ihat_n}(D_n, T_v, U)$ from $\mathsf{def:robust-wcp}$ satisfies $\Pr\{Y^v \text{ in } \mathsf{Ihat_n}(D_n, T_v, U)\} > 1 - \alpha$. Moreover, $\mathsf{Ihat_n}(D_n, T_v, U)$ is a connected interval because it is a finite union of intervals centered at m .*

Proof. Fix an actual candidate $v \in V_\epsilon$, and choose j with $v = v^j$, P_{obs} -almost surely. Candidate normalization gives $\mathbb{E}_{P_{\text{obs}}} v = 1$. By the definition of Q_v , compatible target sampling, and consistency, the observed target triple (T_v, Y^v) has density $v(x, a)$ relative to P_{obs} . Explicitly, for every measurable set $A_0 \subseteq \mathcal{X} \times \mathcal{A} \times [0, 1]$,

$$\Pr\{(T_v, Y^v) \in A_0\} = \mathbb{E}_{P_{\text{obs}}}[v(X, A)\mathbf{1}\{(X, A, Y) \in A_0\}].$$

Pool the n calibration triples and the target triple, independently apply a uniform random permutation, and let J denote the permuted target index. Relative to P_{obs}^{n+1} , the joint density of the permuted pool together with the event $J = l$ is proportional to $v(X_l, A_l)$. Hence, conditional on the pooled triples,

$$\Pr(J = l \mid \text{pooled triples}) = \frac{v(X_l, A_l)}{\sum_{q=1}^{n+1} v(X_q, A_q)}.$$

The denominator is positive almost surely: under the target law,

$$\Pr\{v(T_v) = 0\} = \mathbb{E}_{P_{\text{obs}}}[v(X, A)\mathbf{1}\{v(X, A) = 0\}] = 0.$$

Condition on a pooled realization and write

$$q_l = \frac{v(X_l, A_l)}{\sum_{q=1}^{n+1} v(X_q, A_q)}, \quad s_l = S(Y_l).$$

For the distinguished index J , the candidate- j p-value evaluated at the target outcome is

$$P_J = \sum_{l=1}^{n+1} q_l \mathbf{1}\{s_l > s_J\} + U \sum_{l=1}^{n+1} q_l \mathbf{1}\{s_l = s_J\}.$$

List the distinct score values in decreasing order. If their total q -masses are $a_1, \dots, a_L$, then, conditional on the target score being at level h , P_J is uniform on the interval

$$\left[\sum_{\ell < h} a_\ell, \sum_{\ell \leq h} a_\ell \right].$$

The target level equals h with probability a_h . These adjacent intervals partition $[0, 1]$, so the mixture is uniform on $[0, 1]$. Therefore

$$\Pr\{p_j(Y^v) \leq \alpha\} = \alpha, \quad \Pr\{p_j(Y^v) > \alpha\} = 1 - \alpha.$$

Because $p_{\text{rob}} = \max_k p_k \geq p_j$,

$$\Pr\{Y^v \in \widehat{I}_n(D_n, T_v, U)\} = \Pr\{p_{\text{rob}}(Y^v) > \alpha\} \geq 1 - \alpha.$$

It remains to verify connectedness. For fixed k , a zero denominator makes the candidate acceptance set empty. Otherwise the denominator of $p_k(y)$ is positive and independent of y . As $s = S(y)$ increases, a calibration weight can move only from the strict-upper-tail term, whose coefficient is one, to the tie term, whose coefficient is U , and then to zero. Since $0 < U < 1$ almost surely, the numerator is nonincreasing in s . Thus

$$\{y \in [0, 1] : p_k(y) > \alpha\}$$

is an interval centered at m , possibly open, half-open, or empty. Such centered intervals are nested by their score radii. Their finite union, which is $\widehat{I}_n$, is therefore again a connected interval centered at m . $\square$

Justification. The weighted randomized-rank argument gives exact finite-sample marginal coverage under each candidate-specific target law. *Gap.* Weighted and robust conformal coverage are established baselines; no coverage improvement or simultaneous multi-candidate event is claimed. *Consumer.* Use the nested centered candidate sets to obtain a connected interval, allowing open or half-open endpoints.

Proposition 3 (prop:known-ratio-reduction). *If $K=1$ and $v^1=w$ P_{obs} -almost surely, then $\ell=w=u$, $r_{\text{rob}}=r_w$, and def:robust-wcp equals randomized weighted conformal prediction with ratio w .*

Proof. Assume $K = 1$ and $v^1 = w$ P_{obs} -almost surely. By the definition of the compatible-ratio list,

$$V_\epsilon = \{v^1\}, \quad \ell(x, a) = \min_{1 \leq k \leq 1} v^k(x, a) = v^1(x, a), \quad u(x, a) = \max_{1 \leq k \leq 1} v^k(x, a) = v^1(x, a).$$

Consequently, $\ell = w = u$ P_{obs} -almost surely. Equivalently, because certificate containment requires $w \in V_\epsilon$, the unique ratio element is w ; one may choose the common measurable representative for all three functions.

For every $r \in [0, 1]$, almost-sure equality and boundedness of the indicator give

$$F_{v^1}(r) = \mathbb{E}_{P_{\text{obs}}} [v^1(X, A) \mathbf{1}\{S(Y) \leq r\}] = \mathbb{E}_{P_{\text{obs}}} [w(X, A) \mathbf{1}\{S(Y) \leq r\}] = F_w(r).$$

Since V_ϵ is a singleton, it follows pointwise in r that

$$\inf_{v \in V_\epsilon} F_v(r) = F_{v^1}(r) = F_w(r).$$

The two generalized-quantile sets are therefore identical, and hence

$$r_{\text{rob}} = \inf \left\{ r \in [0, 1] : \inf_{v \in V_\epsilon} F_v(r) \geq 1 - \alpha \right\} = \inf \{ r \in [0, 1] : F_w(r) \geq 1 - \alpha \} = r_w.$$

In particular, the robust oracle interval also reduces to $I_{\text{rob}} = I_w$.

It remains to compare the sample rules. Write $T_w := T_{v^1}$ for the target pair corresponding to the unique candidate. For any candidate outcome $y \in [0, 1]$, the maximum in the robust conformal definition contains only one term, so $p_{\text{rob}}(y) = p_1(y)$. Substituting $v^1 = w$ into that term yields

$$p_1(y) = \frac{\sum_{i=1}^n w(X_i, A_i) \mathbf{1}\{S(Y_i) > S(y)\} + U(w(T_w) + \sum_{i=1}^n w(X_i, A_i) \mathbf{1}\{S(Y_i) = S(y)\})}{w(T_w) + \sum_{i=1}^n w(X_i, A_i)},$$

whenever the denominator is positive, and both definitions assign the value zero when it is not. This is exactly the randomized weighted-conformal p -value $p_w(y)$ for ratio w . Therefore

$$\hat{I}_n = \{y \in [0, 1] : p_{\text{rob}}(y) > \alpha\} = \{y \in [0, 1] : p_w(y) > \alpha\},$$

which proves the claimed reduction. $\square$

Justification. The singleton certified list must recover the known-ratio benchmark. *Gap.* Tibshirani et al. already establish this known-ratio weighted-conformal result; no novelty is claimed for the reduction. *Consumer.* It verifies that a fully logged assignment system can use ordinary weighted conformal prediction.

Lemma 2 (lem:bounded-bernstein-lower-tail). *Let $b > 0$, $V \geq 0$, and let $R_1, \dots, R_n$ be independent centered real random variables satisfying $|R_i| \leq b$ almost surely and $\sum_{i=1}^n \mathbb{E}(R_i^2) \leq V$. Then, for every $t > 0$,*

$$\Pr \left\{ \sum_{i=1}^n R_i \geq t \right\} \leq \exp \left\{ -\frac{t^2}{2(V + bt/3)} \right\}.$$

Consequently, for every $x > 0$, $\sum_{i=1}^n R_i < \sqrt{2Vx} + 2bx/3$ with probability at least $1 - e^{-x}$.

Proof. If $V = 0$, every $R_i = 0$ almost surely and both claims are immediate. Suppose $V > 0$ and fix $0 < \lambda < 3/b$. For every integer $k \geq 2$, $|R_i|^k \leq b^{k-2} R_i^2$. Also $k! \geq 2 \cdot 3^{k-2}$, by induction from $2! = 2$ and $k+1 \geq 3$. Centering and the exponential series therefore give

$$\begin{aligned} \mathbb{E}e^{\lambda R_i} &= 1 + \sum_{k=2}^{\infty} \frac{\lambda^k \mathbb{E}(R_i^k)}{k!} \\ &\leq 1 + \frac{\lambda^2 \mathbb{E}(R_i^2)}{2} \sum_{j=0}^{\infty} \left(\frac{\lambda b}{3} \right)^j \\ &= 1 + \frac{\lambda^2 \mathbb{E}(R_i^2)}{2(1 - \lambda b/3)}. \end{aligned}$$

Using $1 + z \leq e^z$, independence, and $V \geq \sum_i \mathbb{E}(R_i^2)$, we obtain

$$\mathbb{E} \exp \left(\lambda \sum_{i=1}^n R_i \right) \leq \exp \left\{ \frac{\lambda^2 V}{2(1 - \lambda b/3)} \right\}.$$

Markov's inequality gives

$$\Pr \left\{ \sum_i R_i \geq t \right\} \leq \exp \left\{ -\lambda t + \frac{\lambda^2 V}{2(1 - \lambda b/3)} \right\}.$$

Choose $\lambda = t/(V + bt/3)$, which is strictly less than $3/b$ and satisfies $1 - \lambda b/3 = V/(V + bt/3)$. Substitution simplifies the exponent to $-t^2/[2(V + bt/3)]$. Finally, put $t_x = \sqrt{2Vx} + 2bx/3$. Expanding shows $t_x^2 \geq 2x(V + bt_x/3)$, so the first inequality bounds the failure probability at t_x by e^{-x} . $\square$

Lemma 3 (lem:finite-list-heavy-tail-control). *Let $M \in \mathbb{N}$, let $H_1, \dots, H_M$ be measurable nonnegative functions of an observed triple satisfying $H_j(O) \leq u(X, A)$, and write $\mu_j = \mathbb{E}_{P_{\text{obs}}} H_j(O)$. For $\zeta \in (0, 1)$, define $A_\gamma = \gamma/(\gamma - 1)$, $D_\gamma = 1 + 2C/(2 - \gamma)$,*

$$B_{n,\zeta} = \max\{1, (4Cn/\zeta)^{1/\gamma}\}, \quad x_{M,\zeta} = \log(8M/\zeta), \quad T_\zeta = \max\{1, (2KA_\gamma C/\zeta)^{1/(\gamma-1)}\},$$

and

$$d_{n,M,\zeta} = A_\gamma C B_{n,\zeta}^{1-\gamma} + \sqrt{\frac{2D_\gamma B_{n,\zeta}^{2-\gamma} x_{M,\zeta}}{n}} + \frac{2B_{n,\zeta} x_{M,\zeta}}{3n}.$$

Then, for every actual $v \in V_\epsilon$, under the joint law of D_n and T_v , with probability at least $1 - \zeta$,

$$\max_{1 \leq j \leq M} \left| \frac{1}{n} \sum_{i=1}^n H_j(O_i) - \mu_j \right| \leq d_{n,M,\zeta} \quad \text{and} \quad \max_{1 \leq k \leq K} v^k(T_v) \leq T_\zeta.$$

Proof. Write $B = B_{n,\zeta}$, $x = x_{M,\zeta}$, and $A_\gamma = \gamma/(\gamma - 1)$. For every $z \geq 1$, the tail assumption and layer cake give

$$\mathbb{E}_{P_{\text{obs}}}[u \mathbf{1}\{u > z\}] = z \Pr(u > z) + \int_z^\infty \Pr(u > t) dt \leq A_\gamma C z^{1-\gamma}.$$

Also, for $B \geq 1$,

$$\mathbb{E}[(u \wedge B)^2] = 2 \int_0^B t \Pr(u > t) dt \leq 1 + 2C \int_1^B t^{1-\gamma} dt \leq D_\gamma B^{2-\gamma}.$$

The union bound and the definition of B show

$$\Pr \left\{ \max_{1 \leq i \leq n} u(X_i, A_i) > B \right\} \leq nCB^{-\gamma} \leq \zeta/4.$$

For each j , set $\bar{H}_j = H_j \mathbf{1}\{u \leq B\}$. Then $0 \leq \bar{H}_j \leq B$,

$$0 \leq \mu_j - \mathbb{E}\bar{H}_j \leq A_\gamma CB^{1-\gamma}, \quad \mathbb{E}\bar{H}_j^2 \leq D_\gamma B^{2-\gamma}.$$

Apply the bounded Bernstein lemma to the centered variables $\bar{H}_j(O_i) - \mathbb{E}\bar{H}_j$ and to their negatives. A union bound over the resulting $2M$ inequalities gives, with failure probability at most $2Me^{-x} = \zeta/4$,

$$\max_j \left| \frac{1}{n} \sum_{i=1}^n \bar{H}_j(O_i) - \mathbb{E}\bar{H}_j \right| \leq \sqrt{\frac{2D_\gamma B^{2-\gamma} x}{n}} + \frac{2Bx}{3n}.$$

On the event $\max_i u(X_i, A_i) \leq B$, the empirical averages of H_j and $\bar{H}_j$ coincide. Adding the truncation bias proves the first asserted inequality with total failure probability at most $\zeta/2$.

For the target term, fix the actual candidate v . Since $v \leq u$ and $v^k \leq u$, the Q_v change of measure and a union bound give, for every $T \geq 1$,

$$\begin{aligned} \Pr \left\{ \max_k v^k(T_v) > T \right\} &\leq \sum_{k=1}^K \mathbb{E}_{P_{\text{obs}}}[v(X, A) \mathbf{1}\{v^k(X, A) > T\}] \\ &\leq K \mathbb{E}_{P_{\text{obs}}}[u(X, A) \mathbf{1}\{u(X, A) > T\}] \leq KA_\gamma CT^{1-\gamma}. \end{aligned}$$

At $T = T_\zeta$, this is at most $\zeta/2$. Combining the calibration and target events proves the claim. $\square$

Lemma 4 (lem:divisible-support-witnesses). *Let ν denote the dominating measure on $\mathcal{A}$. Define the divisible-intervention-support condition SR_∞ to mean that there exists $x_0 \in \mathcal{X}$ such that, for every sequence $b_0, b_1, \dots \geq 0$ with $\sum_{j \geq 0} b_j = 1$, there is a measurable partition $E_0, E_1, \dots$ of $\mathcal{A}$ satisfying $\int_{E_j} g(a \mid x_0) \nu(da) = b_j$. Under SR_∞ and $K \geq 2$, the following two fixed-certificate witnesses exist. First, for every $\delta \in (0, 1]$, an admissible ordinary-cell certificate satisfies*

$$\sup_{v \in V_\epsilon} \sup_{r \in [0, 1]} |F_v(r) - F_w(r)| \geq \frac{\delta}{2} \min\{1, C(1 + \delta/2)^{-\gamma}\}.$$

Second, for every $\xi > 0$, an admissible fixed two-candidate certificate has actual ordinary-cell width zero and a candidate v for which

$$\mathbb{E}_{P_{\text{obs}}} |v - w| = 2h, \quad \sup_{r \in [0, 1]} |F_v(r) - F_w(r)| = h < \xi,$$

while the supremum over outcome laws sharing that treatment law and certificate of $\text{len}(I_{\text{rob}}) - \text{len}(I_w)$ is one. For all sufficiently small ϵ , the second witness can be chosen with

$$h = \min\{C, 1\} 2^{-\gamma} \epsilon^{\gamma-1} (1 - \epsilon/4),$$

so its CDF discrepancy has the tail order $\epsilon^{\gamma-1}$.

Proof. Write

$$G_0(E) = \int_E g(a \mid x_0) \nu(da)$$

and take P_X degenerate at x_0 .

For the ordinary-cell witness, put

$$d = 1 + \delta/2, \quad p = \frac{1}{2} \min\{1, Cd^{-\gamma}\}.$$

By SR_∞ , choose three treatment cells having G_0 -masses

$$p(1 - \delta/2), \quad p(1 + \delta/2), \quad 1 - 2p.$$

On these cells define

$$w = (1 - \delta/2, 1 + \delta/2, 1), \quad v = (1 + \delta/2, 1 - \delta/2, 1),$$

and define the source density by $\pi = g/w$ cellwise. The source cell probabilities are then $p, p, 1 - 2p$, so π integrates to one. Both w and v normalize under π . Duplicate either candidate if $K > 2$. The envelopes are

$$\ell = (1 - \delta/2, 1 - \delta/2, 1), \quad u = (d, d, 1).$$

Because $d \leq 3/2 < \epsilon^{-1}$, all nonzero widths are ordinary and equal δ . For $1 \leq t < d$,

$$P_{\text{obs}}\{u > t\} = 2p = \min\{1, Cd^{-\gamma}\} \leq Ct^{-\gamma},$$

and the probability is zero for $t \geq d$.

Let $R = \max\{m, 1 - m\}$, and choose $r_0 \in (0, \min\{R, 1/2\})$ sufficiently small. Give the first cell score zero and the other two cells score r_0 , using deterministic potential outcomes. The full potential-outcome vector is then independent of realized treatment, so consistency, exchangeability, and boundedness hold. If the w -mass at score zero is already at least $1 - \alpha$, then $r_w = 0$ and the margin condition is vacuous except at $t = 0$. Otherwise $r_w = r_0$, and r_0 can be chosen so that, for every admissible $t > 0$,

$$F_w(r_0 + t) - F_w(r_0 - t) \geq c_0 t.$$

At radius zero,

$$F_v(0) - F_w(0) = p\delta = \frac{\delta}{2} \min\{1, C(1 + \delta/2)^{-\gamma}\}.$$

This proves the first claim.

For the shell witness, put

$$q = 1 - \alpha, \quad p_0 = q/2, \quad b = \min\{C, 1\}, \quad d_\star = 1/2.$$

Choose $M > \max\{2, \epsilon^{-1}\}$ so large that, with

$$p = \frac{b}{2} M^{-\gamma}, \quad h = p(M - d_\star),$$

one has

$$pM < p_0, \quad pd_\star < \alpha, \quad h < \xi.$$

Set

$$c = \frac{1 - p(M + d_\star)}{1 - 2p}.$$

For sufficiently large M , $0 < c < 1$. Use $\mathbf{SR}_\infty$ to choose five cells E_0, E_1, E_2, E_-, E_+ having G_0 -masses

$$q - p_0, \quad p_0 - pM, \quad \alpha - pd_\star, \quad pM, \quad pd_\star.$$

Define

$$w = (c, c, c, M, d_\star), \quad v = (c, c, c, d_\star, M)$$

on these cells and let $\pi = g/w$ cellwise. The source masses of E_- and E_+ are both p , and the total source mass of the first three cells is

$$\frac{1 - p(M + d_\star)}{c} = 1 - 2p.$$

Thus π , w , and v are normalized. The envelopes equal c on the first three cells and satisfy $\ell = d_\star$, $u = M$ on the two shell cells. Hence the actual ordinary width is zero, while on the shell

$$u - \ell = M - d_\star \leq u$$

and $u > \epsilon^{-1}$. Moreover, for $1 \leq t < M$,

$$P_{\text{obs}}\{u > t\} = 2p = bM^{-\gamma} \leq Ct^{-\gamma},$$

and the tail is zero for $t \geq M$.

Choose $r_0 > 0$ with

$$2r_0 < R, \quad r_0 \leq p_0/c_0.$$

Assign scores

$$(0, r_0, R, r_0, R)$$

on $(E_0, E_1, E_2, E_-, E_+)$, again using deterministic potential outcomes. Under the actual intervention ratio w , the score masses at $0, r_0, R$ are

$$q - p_0, \quad p_0, \quad \alpha.$$

Under v , they are

$$q - p_0, \quad p_0 - h, \quad \alpha + h.$$

Consequently,

$$r_w = r_0, \quad r_v = r_{\text{rob}} = R.$$

For every admissible $t > 0$, $t \leq r_0$ and

$$F_w(r_0 + t) - F_w(r_0 - t) = p_0 \geq c_0 t,$$

so the frozen symmetric margin holds. Between r_0 and R , the two CDFs differ by exactly h , and elsewhere they agree. Also,

$$\mathbb{E}|v - w| = 2p(M - d_*) = 2h.$$

The robust interval is $[0, 1]$, whereas $\text{len}(I_w) \leq 2r_0$. The treatment law and certificate do not depend on r_0 ; letting $r_0 \downarrow 0$ across outcome laws sharing those objects proves that the supremum of the length price is one.

Finally take $M = 2\epsilon^{-1}$. For all sufficiently small ϵ , the preceding inequalities hold and

$$h = \frac{b}{2}M^{-\gamma}(M - d_*) = b2^{-\gamma}\epsilon^{\gamma-1}(1 - \epsilon/4).$$

This proves the tail-order assertion. $\square$

Theorem 1 (thm:population-length-frontier-degeneracy). *For every fixed admissible certificate,*

$$\sup_{v \in V_\epsilon} \sup_{r \in [0, 1]} |F_v(r) - F_w(r)| \leq \min\{1, K - 1, H(C, \gamma), W/2\},$$

with $H(C, \gamma)$ and W as in *lem:certificate-cdf-modulus*, and

$$0 \leq \text{len}(I_{\text{rob}}) - \text{len}(I_w) \leq 1.$$

If $K = 1$, the length difference is identically zero. For $K \geq 2$, its supremum is not determined by $(\delta, \epsilon, \gamma, C, K)$ alone: duplicate candidate lists have zero price, while under the divisible-intervention-support condition SR_∞ of **lem:divisible-support-witnesses**, there are fixed two-candidate certificates for which

$$\sup_{P_F \in \mathfrak{P}_{\text{cert}}} \{\text{len}(I_{\text{rob}}) - \text{len}(I_w)\} = 1.$$

Thus, under SR_∞ , the worst case over admissible support-rich fixed certificates has sharp order one, even with zero actual ordinary-cell width and arbitrarily small CDF discrepancy. Separately, admissible certificates attain ordinary-cell CDF discrepancy of order δ and tail-shell CDF discrepancy of order $\epsilon^{\gamma-1}$. Merely attaining those CDF moduli is not sufficient for a connected-length rate. A one-sided upper-quantile growth condition would yield the desired inversion, but that condition is absent from the frozen assumptions.

Proof. The first display is exactly **lem:certificate-cdf-modulus**. Because certificate containment gives $w \in V_\epsilon$, for every r

$$\inf_{v \in V_\epsilon} F_v(r) \leq F_w(r).$$

Therefore the first radius at which the left side reaches $1 - \alpha$ cannot be smaller than the corresponding w -radius:

$$r_{\text{rob}} \geq r_w.$$

The map

$$L_m(r) = \text{len}([m - r, m + r] \cap [0, 1])$$

is nondecreasing and takes values in $[0, 1]$. Hence

$$0 \leq L_m(r_{\text{rob}}) - L_m(r_w) \leq 1.$$

If $K = 1$, containment forces the sole candidate to equal w , P_{obs} -almost surely. The two defining CDFs and radii then coincide, so the price is zero. The same conclusion holds for any K when every listed candidate is a duplicate of w . Such a zero-price certificate exists for the same numerical parameters: take $\pi = g$, set every candidate equal to $w = 1$, and take $Y(a) = m$ almost surely. Then $u = \ell = 1$, the strict upper tail above every $t \geq 1$ is zero, the width is zero, and $r_w = 0$, so the frozen symmetric margin is vacuous except at $t = 0$.

Suppose now that $K \geq 2$ and SR_∞ holds. The fixed shell certificate constructed in **lem:divisible-support-witnesses** satisfies all frozen restrictions, has actual ordinary-cell width zero, and keeps its treatment law and candidate list fixed while only the outcome law varies. For that one fixed certificate the lemma proves

$$\sup_{P_F \in \mathfrak{P}_{\text{cert}}} \{\text{len}(I_{\text{rob}}) - \text{len}(I_w)\} = 1.$$

The universal unit upper bound proves sharpness. Since a duplicate K -candidate certificate has price zero with the same numerical class parameters, no function of $(\delta, \epsilon, \gamma, C, K)$ alone can equal the fixed-certificate supremum.

The same lemma gives an ordinary-cell construction with CDF discrepancy

$$\frac{\delta}{2} \min\{1, C(1 + \delta/2)^{-\gamma}\},$$

which is of order δ for fixed $C > 0$, and a shell construction whose discrepancy for small ϵ is

$$\min\{C, 1\} 2^{-\gamma} \epsilon^{\gamma-1} (1 - \epsilon/4).$$

Thus both componentwise CDF exponents are genuine. Nevertheless the shell construction allows this discrepancy to be arbitrarily small while the length-price supremum remains one.

The obstruction is quantile inversion. The frozen margin controls

$$F_w(r_w + t) - F_w(r_w - t),$$

but permits all of that increment to lie weakly to the left of r_w , and at a boundary quantile it can be one-sidedly vacuous. It therefore permits a flat interval immediately to the right of r_w . If one additionally imposed

$$F_w(r_w + t) - F_w(r_w) \geq c_+ t$$

for the relevant right-hand range, then a uniform CDF discrepancy Δ would imply

$$r_{\text{rob}} - r_w \leq \Delta/c_+$$

whenever the endpoint lies in $[0, 1]$, and the length difference would be at most $2\Delta/c_+$. That one-sided condition is not frozen. Hence shell richness that only attains the CDF modulus cannot produce the requested vanishing connected-length frontier. $\square$

Theorem 2 (thm:sample-regret-degeneracy). *The frozen marginal-coverage min-max regret is degenerate. For every n ,*

$$0 \leq R_n(\eta) \leq 1,$$

and if $\eta \geq 1 - \alpha$, then $R_n(\eta) = 0$ for every n . For arbitrary $\eta \in (0, 1)$, put $\zeta = \eta/2$, $A_\gamma = \gamma/(\gamma - 1)$, $D_\gamma = 1 + 2C/(2 - \gamma)$, and

$$B_{n,\zeta} = \max\{1, (4Cn/\zeta)^{1/\gamma}\}, \quad x_\zeta = \log(16K/\zeta), \quad T_\zeta = \max\{1, (2KA_\gamma C/\zeta)^{1/(\gamma-1)}\},$$

$$d_{n,\zeta} = A_\gamma C B_{n,\zeta}^{1-\gamma} + \sqrt{\frac{2D_\gamma B_{n,\zeta}^{2-\gamma} x_\zeta}{n} + \frac{2B_{n,\zeta} x_\zeta}{3n}}, \quad q_{n,\zeta} = \frac{(1 + \alpha)d_{n,\zeta} + T_\zeta/n}{1 - d_{n,\zeta}}.$$

Whenever

$$d_{n,\zeta} < 1, \quad q_{n,\zeta} < 1 - \alpha, \quad \frac{q_{n,\zeta}}{\alpha + q_{n,\zeta}} \leq \eta/2,$$

one has the exact identity $R_n(\eta) = 0$. These conditions hold for all sufficiently large n , because

$$q_{n,\zeta} = O_{C,\gamma,K,\eta,\alpha}(n^{-(\gamma-1)/\gamma}).$$

An attaining rule is a measurable closed score-centered hull of robust weighted conformal prediction run at miscoverage $\alpha + q_{n,\zeta}$, mixed with $[0, 1]$ with probability $q_{n,\zeta}/(\alpha + q_{n,\zeta})$. It is marginally $1 - \alpha$ valid and uses no second moment. Thus $n^{-(\gamma-1)/\gamma}$ is construction slack, not a positive regret rate.

Moreover, under SR_∞ , for every $\eta \in (0, 1)$ there is a single fixed normalized duplicate-candidate certificate and a single fixed source treatment law satisfying the frozen tail and width restrictions such that, along a subsequence n_j , admissible outcome laws with those same treatment objects satisfy

$$\Pr\{\text{len}(\widehat{I}_{n_j}) - \text{len}(I_{\text{rob}}) \geq \Delta\} > \eta$$

for some fixed $\Delta > 0$. Hence the unmodified exact- α rule need not attain the zero minimax frontier; the displayed randomized repair does attain it under the stated finite-sample conditions.

Proof. The lower bound $R_n(\eta) \geq 0$ is part of the definition. The rule that always returns $[0, 1]$ is score-centered, has coverage one, and has excess length at most one. Hence $R_n(\eta) \leq 1$.

If $\eta \geq 1 - \alpha$, use U to return $[0, 1]$ with probability $1 - \alpha$ and $\{m\}$ otherwise. This rule has coverage at least $1 - \alpha$ under every candidate target law. Positive excess length can occur only on the full-interval branch, whose probability is at most η . Thus zero is feasible in the inner regret quantile and $R_n(\eta) = 0$.

Now fix arbitrary $\eta \in (0, 1)$ and put $\zeta = \eta/2$. Split U , measurably, into two independent uniforms U_1, U_2 by assigning the odd binary digits of its nonterminating expansion to U_1 and the even digits to U_2 . The dyadic ambiguity is a null event. Thus U_1, U_2 remain jointly independent of the sample and target.

For a fixed law define

$$G(r) = \min_{1 \leq k \leq K} F_{v^k}(r).$$

Candidate normalization makes each F_{v^k} a CDF. The finite minimum G is nondecreasing and right-continuous, and $G(1) = 1$. Right continuity at the infimum defining r_{rob} therefore yields

$$F_{v^k}(r_{\text{rob}}) \geq 1 - \alpha \quad \text{for every } k.$$

Equivalently,

$$\mu_k := \mathbb{E}_{P_{\text{obs}}}[v^k(X, A)\mathbf{1}\{S(Y) > r_{\text{rob}}\}] \leq \alpha.$$

Apply `lem:finite-list-heavy-tail-control` with the $2K$ functions

$$v^k, \quad v^k\mathbf{1}\{S > r_{\text{rob}}\}, \quad 1 \leq k \leq K.$$

For every actual target candidate v , with joint probability at least $1 - \zeta$, simultaneously in k ,

$$\frac{1}{n} \sum_{i=1}^n v^k(X_i, A_i) \geq 1 - d_{n,\zeta},$$

$$\frac{1}{n} \sum_{i=1}^n v^k(X_i, A_i) \mathbf{1}\{S(Y_i) > r_{\text{rob}}\} \leq \alpha + d_{n,\zeta}, \quad v^k(T_v) \leq T_\zeta.$$

Call this event $\mathcal{E}_v$. For any y with $S(y) > r_{\text{rob}}$, the numerator of $p_k(y)$, computed with U_1 , is at most

$$\sum_{i=1}^n v^k(X_i, A_i) \mathbf{1}\{S(Y_i) > r_{\text{rob}}\} + v^k(T_v).$$

On $\mathcal{E}_v$,

$$p_k(y) \leq \frac{\alpha + d_{n,\zeta} + T_\zeta/n}{1 - d_{n,\zeta}} = \alpha + q_{n,\zeta}.$$

Define the measurable radius

$$\tilde{h}_n = \sup \left(\{S(y) : y \in \mathbb{Q} \cap [0, 1], p_{\text{rob}}^{(U_1)}(y) > \alpha + q_{n,\zeta}\} \cup \{0\} \right)$$

and the closed score-centered interval

$$\tilde{J}_n = [m - \tilde{h}_n, m + \tilde{h}_n] \cap [0, 1].$$

Each rational-point p-value is measurable, so $\tilde{h}_n$ is measurable. Monotonicity of every candidate p-value in the score implies that the robust acceptance set is downward closed in S . Density of the rationals then shows that $\tilde{J}_n$ is its closed centered hull, with $\{m\}$ adjoined if it is empty. It contains the acceptance set. On $\mathcal{E}_v$,

$$\tilde{h}_n \leq r_{\text{rob}}, \quad \tilde{J}_n \subseteq I_{\text{rob}},$$

so its excess length is nonpositive.

Apply `prop:robust-wcp-coverage` with level

$$\alpha' = \alpha + q_{n,\zeta} < 1$$

and randomizer U_1 . Taking the closed hull can only increase coverage, hence

$$\Pr\{Y^v \in \tilde{J}_n\} \geq 1 - \alpha - q_{n,\zeta}.$$

Set

$$\lambda = \frac{q_{n,\zeta}}{\alpha + q_{n,\zeta}}$$

and define $J_n^\star = [0, 1]$ when $U_2 \leq \lambda$, and $J_n^\star = \tilde{J}_n$ otherwise. Independence gives

$$\Pr\{Y^v \in J_n^\star\} \geq \lambda + (1 - \lambda)(1 - \alpha - q_{n,\zeta})$$

$$= 1 - \alpha - q_{n,\zeta} + \lambda(\alpha + q_{n,\zeta}) = 1 - \alpha.$$

Thus $J_n^\star$ is feasible in `def:minimax-regret`. Positive excess length can occur only on the full-interval branch or on $\mathcal{E}_v^c$, so, uniformly over laws and candidates,

$$\Pr\{\text{len}(J_n^\star) - \text{len}(I_{\text{rob}}) > 0\} \leq \lambda + \zeta \leq \eta.$$

Therefore $R_n(\eta) = 0$.

Since $B_{n,\zeta} \asymp n^{1/\gamma}$, each of the three terms in $d_{n,\zeta}$ is

$$O_{C,\gamma,K,\eta}(n^{-(\gamma-1)/\gamma}),$$

and $T_\zeta/n = O(n^{-1})$. Hence the displayed finite- n conditions hold eventually. No second moment was used.

Before constructing the counterexample, record the exact-level hull identity for possibly open acceptance intervals. Let p_{rob}^α denote the robust p-value used with the exact threshold α , and define

$$J_0 = \{y \in [0, 1] : p_{\text{rob}}^\alpha(y) > \alpha\} = \widehat{I}_n.$$

Define its measurable countable-rational score supremum by

$$h_0 = \sup(\{S(y) : y \in \mathbb{Q} \cap [0, 1], p_{\text{rob}}^\alpha(y) > \alpha\} \cup \{0\})$$

and its closed centered hull by

$$\bar{J}_0 = [m - h_0, m + h_0] \cap [0, 1].$$

Measurability of h_0 follows because it is the supremum of countably many measurable threshold indicators. By the score monotonicity proved in **prop:robust-wcp-coverage**, J_0 is a centered interval downward closed in S . If $J_0 \neq \emptyset$, density of $\mathbb{Q}$ in every nondegenerate interval and continuity of $S(y) = |y - m|$ show that $h_0 = \sup_{y \in J_0} S(y)$. The only possible difference between J_0 and $\bar{J}_0$ is then one or two endpoints, including the open and half-open cases. Hence they have equal one-dimensional Lebesgue measure. If $J_0 = \emptyset$, the rational set is empty, $h_0 = 0$, and $\bar{J}_0 = \{m\}$; adjoining this singleton changes neither Lebesgue measure nor coverage adversely. In every case,

$$\lambda_1(J_0) = \lambda_1(\bar{J}_0) = \text{len}(\bar{J}_0).$$

It remains to prove the claim about the unmodified exact-level rule. Assume SR_∞ , fix $\eta \in (0, 1)$, and choose

$$0 < c < \min\{C/4, -\log \eta\}.$$

Because $1 - \gamma/2 > 0$, one can choose recursively an increasing integer sequence n_j and put $M_j = n_j^{1/\gamma}$ so that

$$\begin{aligned} \sum_{\ell \geq j} M_\ell^{-\gamma} &\leq 2M_j^{-\gamma}, & n_j \sum_{\ell > j} M_\ell^{-\gamma} &\longrightarrow 0, \\ \sum_{\ell > j} M_\ell^{1-\gamma} &\longrightarrow 0, & \frac{M_{j-1}}{M_j^{1-\gamma/2}} &\longrightarrow 0. \end{aligned}$$

At each recursion step these requirements involve only finitely many earlier choices and are met by taking the next term sufficiently large.

Put

$$p_j = cM_j^{-\gamma} = \frac{c}{n_j}, \quad h_j = p_j M_j = cM_j^{1-\gamma}.$$

Choose M_1 large enough that

$$P := \sum_j p_j < 1, \quad H := \sum_j h_j < 1.$$

By SR_∞ , partition the intervention law at x_0 into a baseline cell E_0 of mass $1 - H$ and shell cells E_j of masses h_j . Define

$$w = M_j \quad \text{on } E_j, \quad w = w_0 := \frac{1 - H}{1 - P} \quad \text{on } E_0,$$

and define $\pi = g/w$ cellwise. The source masses are p_j on E_j and $1 - P$ on E_0 , so π is normalized. Since $H > P$, $0 < w_0 < 1$. Take every candidate equal to this one fixed w . Then $\ell = u = w$, containment and normalization hold, and the certificate width is zero.

For $M_{j-1} \leq t < M_j$, with $M_0 = 1$,

$$P_{\text{obs}}\{u > t\} = \sum_{\ell \geq j} p_\ell \leq 2cM_j^{-\gamma} \leq Ct^{-\gamma}.$$

Thus the same fixed treatment law and certificate satisfy the frozen tail envelope.

Choose fixed radii

$$0 < 2r_0 < r_1 < R := \max\{m, 1 - m\}, \quad r_0 < \frac{1 - \alpha}{2c_0},$$

and set

$$\Delta = \text{len}([m - r_1, m + r_1] \cap [0, 1]) - \text{len}([m - r_0, m + r_0] \cap [0, 1]) > 0.$$

For sample size n_j , define an outcome law as follows. On E_j , the score is r_0 . Off E_j , using latent randomness independent of treatment, the score is uniform on $[0, r_0]$ with probability

$$\frac{1 - \alpha - h_j}{1 - h_j}$$

and is r_1 with probability

$$\beta_j := \frac{\alpha}{1 - h_j}.$$

For all large j these are valid probabilities. Realize the scores by moving from m toward its farther endpoint. The potential-outcome vector is generated independently of realized treatment, so consistency, exchangeability, and boundedness hold.

Under the intervention law, the score distribution has mass $1 - \alpha - h_j$ uniform on $[0, r_0]$, mass h_j at r_0 , and mass α at r_1 . Therefore

$$r_w = r_{\text{rob}} = r_0.$$

For every admissible t ,

$$F_w(r_0 + t) - F_w(r_0 - t) = h_j + \frac{1 - \alpha - h_j}{r_0} t \geq c_0 t$$

for all sufficiently large j . Hence each outcome law belongs to the same fixed-certificate class.

Let $\mathcal{A}_j$ be the event that neither the calibration sample nor the target pair falls in any shell E_ℓ with $\ell \geq j$. The source and target probabilities and the growth conditions give

$$\Pr(\mathcal{A}_j) = \left(1 - \sum_{\ell \geq j} p_\ell\right)^{n_j} \left(1 - \sum_{\ell \geq j} h_\ell\right) \rightarrow e^{-c} > \eta.$$

On $\mathcal{A}_j$, every calibration and target weight is at most

$$b_j = \max\{w_0, M_{j-1}\}.$$

For any candidate score $s \in (r_0, r_1)$, let B_i indicate that calibration score i equals r_1 , let $W_i = w(X_i, A_i)$, and let $W_T = w(T_w)$. Conditional on the treatment draws and $\mathcal{A}_j$, the B_i are independent Bernoulli variables with mean β_j , independent of the weights, and

$$p_{\text{rob}}(s) > \alpha \iff \sum_{i=1}^{n_j} W_i (B_i - \alpha) + (U - \alpha) W_T > 0.$$

The deterministic conditional drift satisfies

$$(\beta_j - \alpha) \sum_{i=1}^{n_j} W_i \geq \frac{\alpha h_j}{1 - h_j} n_j w_0 = \frac{\alpha c w_0}{1 - h_j} M_j.$$

The centered sum has conditional variance at most $n_j b_j^2$, while $|(U - \alpha) W_T| \leq b_j$. The recursive growth condition gives

$$\frac{n_j b_j^2}{M_j^2} = b_j^2 M_j^{\gamma-2} \rightarrow 0, \quad \frac{b_j}{M_j} \rightarrow 0.$$

Chebyshev's inequality therefore shows, uniformly over the treatment draws in $\mathcal{A}_j$, that the last displayed strict inequality holds with conditional probability tending to one.

Consequently, with probability tending to at least e^{-c} , the exact- α acceptance set $J_0 = \hat{I}_{n_j}$ contains every score in (r_0, r_1) . Score monotonicity makes J_0 contain the centered interval of radius r_1 , up to endpoints of zero measure. The established identity $\lambda_1(J_0) = \lambda_1(\bar{J}_0) = \text{len}(\bar{J}_0)$ therefore transfers this lower bound unchanged to the measurable closed centered hull. Hence

$$\liminf_{j \rightarrow \infty} \Pr\{\text{len}(\hat{I}_{n_j}) - \text{len}(I_{\text{rob}}) \geq \Delta\} \geq e^{-c} > \eta.$$

The source treatment law and duplicate candidate list are fixed across j ; only the admissible outcome law varies, as permitted by the supremum in the minimax criterion. This proves that the unmodified exact-level rule is not a uniform zero-regret witness. $\square$

7 Interpretation

Coverage is finite-sample marginal coverage evaluated separately under each candidate-specific target law, not a simultaneous event across candidates. Candidate acceptance sets are score-centered and nested, so their finite union is a connected interval that may be open or half-open. For the exact-level set $J_0 = \{y : p_{\text{rob}}^\alpha(y) > \alpha\}$, the countable-rational score supremum h_0 is measurable and defines $\bar{J}_0 = [m - h_0, m + h_0] \cap [0, 1]$. Downward-closedness in score and rational density give $\lambda_1(J_0) = \lambda_1(\bar{J}_0) = \text{len}(\bar{J}_0)$; if J_0 is empty, adjoining $\{m\}$ changes neither measure nor coverage adversely. Population conclusions must be read in two layers: the unit upper bound is universal, whereas the supremum-one lower construction is conditional on SR_∞ , or equivalently is a worst case over admissible support-rich certificates.

Technical internal limitation (diagnostic only)

For every measurable H with $0 \leq H \leq 1$, normalization gives $|\mathbb{E}[(v - w)H]| \leq \min\{\mathbb{E}(u - w), \mathbb{E}(w - \ell), \mathbb{E}(u - \ell)/2\}$. Tail integration and graded width yield $\min\{1, K - 1, H(C, \gamma), W/2\}$, but no joint sharpness of this full minimum is claimed. The frozen symmetric quantile margin permits a flat interval immediately to the right of r_w , so vanishing CDF discrepancy need not imply vanishing connected length. The order-one lower mechanism additionally requires the witness-side support divisibility SR_∞ ; without it, only the universal unit upper bound is established. Separately, marginal validity plus an objective ignoring an η -probability event permits a measurable randomized repair, making $R_n(\eta)$ eventually zero. The heavy-tail scale $n^{-(\gamma-1)/\gamma}$ controls the repair slack and is not a minimax regret rate.

8 Honest scope

All conclusions are conditional on a fixed externally supplied finite candidate list with exact containment, conditional normalization, graded width, and the stated upper-tail envelope. The note proves no list-learning, estimation, auditing, or certification procedure. It gives candidate-wise marginal coverage rather than joint multi-candidate coverage. Universally, the population connected-length price lies in $[0, 1]$, and it is zero for an effectively singleton list. For $K \geq 2$, the supremum-one lower conclusion holds only under the formally defined divisible-intervention-support condition SR_∞ , hence as a worst case over admissible support-rich certificates; it is not asserted for every support. The marginal regret has an all-sample-size zero branch when $\eta \geq 1 - \alpha$ and is eventually zero below that elbow, so the original positive heavy-tail regret frontier is not identified by this criterion.

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